

Table S4: Gene ontologies enriched in Ning *et al* dataset differentiating GOLD 0 from GOLD2 smokers.

GOID	Ontology	Term	p
GO:0006414	biological_process	translational elongation	1.43E-24
GO:0006412	biological_process	translation	2.89E-19
GO:0043284	biological_process	biopolymer biosynthetic process	1.11E-11
GO:0034961	biological_process	cellular biopolymer biosynthetic process	3.21E-11
GO:0009059	biological_process	macromolecule biosynthetic process	1.01E-10
GO:0009058	biological_process	biosynthetic process	3.86E-10
GO:0034645	biological_process	cellular macromolecule biosynthetic process	2.67E-09
GO:0044249	biological_process	cellular biosynthetic process	4.75E-09
GO:0009987	biological_process	cellular process	5.78E-09
GO:0010467	biological_process	gene expression	7.06E-09
GO:0044237	biological_process	cellular metabolic process	7.60E-08
GO:0008152	biological_process	metabolic process	2.18E-07
GO:0043170	biological_process	macromolecule metabolic process	2.18E-07
GO:0044260	biological_process	cellular macromolecule metabolic process	1.44E-06
GO:0044238	biological_process	primary metabolic process	1.77E-06
GO:0019538	biological_process	protein metabolic process	2.47E-06
GO:0044267	biological_process	cellular protein metabolic process	7.74E-06
GO:0010035	biological_process	response to inorganic substance	8.88E-06
GO:0043283	biological_process	biopolymer metabolic process	3.69E-05
GO:0006172	biological_process	ADP biosynthetic process	4.18E-05
GO:0009136	biological_process	purine nucleoside diphosphate biosynthetic process	4.18E-05
GO:0009180	biological_process	purine ribonucleoside diphosphate biosynthetic process	4.18E-05
GO:0009188	biological_process	ribonucleoside diphosphate biosynthetic process	4.18E-05
GO:0016043	biological_process	cellular component organization	4.79E-05
GO:0042274	biological_process	ribosomal small subunit biogenesis	4.99E-05
GO:0044085	biological_process	cellular component biogenesis	7.38E-05
GO:0046031	biological_process	ADP metabolic process	0.000116

GO:0034960	biological_process	cellular biopolymer metabolic process	0.000173
GO:0009133	biological_process	nucleoside diphosphate biosynthetic process	0.000256
GO:0009135	biological_process	purine nucleoside diphosphate metabolic process	0.000256
GO:0009179	biological_process	purine ribonucleoside diphosphate metabolic process	0.000256
GO:0030155	biological_process	regulation of cell adhesion	0.000383
GO:0022613	biological_process	ribonucleoprotein complex biogenesis	0.000383
GO:0016192	biological_process	vesicle-mediated transport	0.00079
GO:0033003	biological_process	regulation of mast cell activation	0.000823
GO:0009185	biological_process	ribonucleoside diphosphate metabolic process	0.000823
GO:0060325	biological_process	face morphogenesis	0.000823
GO:0002886	biological_process	regulation of myeloid leukocyte mediated immunity	0.000823
GO:0044092	biological_process	negative regulation of molecular function	0.00111
GO:0006364	biological_process	rRNA processing	0.00111
GO:0042273	biological_process	ribosomal large subunit biogenesis	0.001302
GO:0002697	biological_process	regulation of immune effector process	0.001398
GO:0016072	biological_process	rRNA metabolic process	0.001504
GO:0042254	biological_process	ribosome biogenesis	0.001709
GO:0048545	biological_process	response to steroid hormone stimulus	0.001741
GO:0000302	biological_process	response to reactive oxygen species	0.001791
GO:0010926	biological_process	anatomical structure formation	0.001797
GO:0030260	biological_process	entry into host cell	0.001803
GO:0044409	biological_process	entry into host	0.001803
GO:0046718	biological_process	entry of virus into host cell	0.001803
GO:0051806	biological_process	entry into cell of other organism during symbiotic interaction	0.001803
GO:0051828	biological_process	entry into other organism during symbiotic interaction	0.001803
GO:0052126	biological_process	movement in host environment	0.001803
GO:0052192	biological_process	movement in environment of other organism during symbiotic interaction	0.001803

GO:0009725	biological process	response to hormone stimulus	0.001803
GO:0060323	biological process	head morphogenesis	0.001803
GO:0060324	biological process	face development	0.001803
GO:0051179	biological process	localization	0.002248
GO:0010033	biological process	response to organic substance	0.002379
GO:0042221	biological process	response to chemical stimulus	0.00244
GO:0007162	biological process	negative regulation of cell adhesion	0.002479
GO:0016044	biological process	membrane organization	0.002644
GO:0010038	biological process	response to metal ion	0.002981
GO:0002246	biological process	healing during inflammatory response	0.00383
GO:0032964	biological process	collagen biosynthetic process	0.00383
GO:0006787	biological process	porphyrin catabolic process	0.00383
GO:0033015	biological process	tetrapyrrole catabolic process	0.00383
GO:0044275	biological process	cellular carbohydrate catabolic process	0.004204
GO:0032410	biological process	negative regulation of transporter activity	0.004771
GO:0010171	biological process	body morphogenesis	0.004771
GO:0060322	biological process	head development	0.004771
GO:0009719	biological process	response to endogenous stimulus	0.005017
GO:0009605	biological process	response to external stimulus	0.00507
GO:0006091	biological process	generation of precursor metabolites and energy	0.00538
GO:0006200	biological process	ATP catabolic process	0.00623
GO:0006901	biological process	vesicle coating	0.00623
GO:0009612	biological process	response to mechanical stimulus	0.00664
GO:0019062	biological process	virion attachment to host cell surface receptor	0.007026
GO:0030198	biological process	extracellular matrix organization	0.007472
GO:0030837	biological process	negative regulation of actin filament polymerization	0.007562
GO:0009132	biological process	nucleoside diphosphate metabolic process	0.007963
GO:0006900	biological process	membrane budding	0.007963
GO:0030833	biological process	regulation of actin filament polymerization	0.008187
GO:0032272	biological process	negative regulation of protein polymerization	0.008732
GO:0001568	biological process	blood vessel development	0.009191

GO:0030036	biological_process	actin cytoskeleton organization	0.009191
GO:0022607	biological_process	cellular component assembly	0.010464
GO:0051641	biological_process	cellular localization	0.011145
GO:0033627	biological_process	cell adhesion mediated by integrin	0.01123
GO:0043304	biological_process	regulation of mast cell degranulation	0.01123
GO:0032365	biological_process	intracellular lipid transport	0.01123
GO:0032409	biological_process	regulation of transporter activity	0.01129
GO:0001944	biological_process	vasculature development	0.011323
GO:0006007	biological_process	glucose catabolic process	0.011965
GO:0009203	biological_process	ribonucleoside triphosphate catabolic process	0.012045
GO:0009207	biological_process	purine ribonucleoside triphosphate catabolic process	0.012045
GO:0009611	biological_process	response to wounding	0.012322
GO:0006950	biological_process	response to stress	0.013211
GO:0002703	biological_process	regulation of leukocyte mediated immunity	0.014215
GO:0051494	biological_process	negative regulation of cytoskeleton organization	0.014215
GO:0034660	biological_process	ncRNA metabolic process	0.014411
GO:0051701	biological_process	interaction with host	0.014866
GO:0051702	biological_process	interaction with symbiont	0.014866
GO:0048519	biological_process	negative regulation of biological process	0.014866
GO:0006810	biological_process	transport	0.014866
GO:0046907	biological_process	intracellular transport	0.014866
GO:0009146	biological_process	purine nucleoside triphosphate catabolic process	0.014866
GO:0010714	biological_process	positive regulation of collagen metabolic process	0.014866
GO:0032967	biological_process	positive regulation of collagen biosynthetic process	0.014866
GO:0051271	biological_process	negative regulation of cellular component movement	0.014866
GO:0030029	biological_process	actin filament-based process	0.014866

GO:0008064	biological_process	regulation of actin polymerization or depolymerization	0.014866
GO:0030832	biological_process	regulation of actin filament length	0.014866
GO:0031333	biological_process	negative regulation of protein complex assembly	0.014866
GO:0032271	biological_process	regulation of protein polymerization	0.014866
GO:0032414	biological_process	positive regulation of ion transmembrane transporter activity	0.014866
GO:0042542	biological_process	response to hydrogen peroxide	0.014866
GO:0051591	biological_process	response to cAMP	0.014866
GO:0055002	biological_process	striated muscle cell development	0.014866
GO:0032413	biological_process	negative regulation of ion transmembrane transporter activity	0.014866
GO:0060314	biological_process	regulation of ryanodine-sensitive calcium-release channel activity	0.014866
GO:0034599	biological_process	cellular response to oxidative stress	0.014866
GO:0043300	biological_process	regulation of leukocyte degranulation	0.014866
GO:0001935	biological_process	endothelial cell proliferation	0.014866
GO:0010639	biological_process	negative regulation of organelle organization	0.015499
GO:0002714	biological_process	positive regulation of B cell mediated immunity	0.016143
GO:0002891	biological_process	positive regulation of immunoglobulin mediated immune response	0.016143
GO:0042060	biological_process	wound healing	0.016143
GO:0001558	biological_process	regulation of cell growth	0.016143
GO:0006098	biological_process	pentose-phosphate shunt	0.016143
GO:0043537	biological_process	negative regulation of blood vessel endothelial cell migration	0.016143
GO:0055093	biological_process	response to hyperoxia	0.016143
GO:0002712	biological_process	regulation of B cell mediated immunity	0.016948
GO:0002889	biological_process	regulation of immunoglobulin mediated immune response	0.016948
GO:0019059	biological_process	initiation of viral infection	0.016948
GO:0009154	biological_process	purine ribonucleotide catabolic process	0.016948
GO:0022407	biological_process	regulation of cell-cell adhesion	0.016948

GO:0051693	biological_process	actin filament capping	0.016948
GO:0043242	biological_process	negative regulation of protein complex disassembly	0.017281
GO:0050776	biological_process	regulation of immune response	0.017461
GO:0051234	biological_process	establishment of localization	0.017669
GO:0051129	biological_process	negative regulation of cellular component organization	0.018238
GO:0019320	biological_process	hexose catabolic process	0.018795
GO:0044403	biological_process	symbiosis, encompassing mutualism through parasitism	0.019018
GO:0043062	biological_process	extracellular structure organization	0.019018
GO:0015985	biological_process	energy coupled proton transport, down electrochemical gradient	0.019018
GO:0015986	biological_process	ATP synthesis coupled proton transport	0.019018
GO:0009143	biological_process	nucleoside triphosphate catabolic process	0.019553
GO:0006754	biological_process	ATP biosynthetic process	0.020257
GO:0051101	biological_process	regulation of DNA binding	0.020621
GO:0051270	biological_process	regulation of cellular component movement	0.020799
GO:0006740	biological_process	NADPH regeneration	0.020799
GO:0032411	biological_process	positive regulation of transporter activity	0.020799
GO:0043433	biological_process	negative regulation of transcription factor activity	0.020799
GO:0090048	biological_process	negative regulation of transcription regulator activity	0.020799
GO:0010812	biological_process	negative regulation of cell-substrate adhesion	0.020799
GO:0048194	biological_process	Golgi vesicle budding	0.020799
GO:0048200	biological_process	Golgi transport vesicle coating	0.020799
GO:0048205	biological_process	COPI coating of Golgi vesicle	0.020799
GO:0007010	biological_process	cytoskeleton organization	0.020969
GO:0046365	biological_process	monosaccharide catabolic process	0.021004
GO:0065003	biological_process	macromolecular complex assembly	0.021198
GO:0008104	biological_process	protein localization	0.021418
GO:0009261	biological_process	ribonucleotide catabolic process	0.021862

GO:0050777	biological_process	negative regulation of immune response	0.021862
GO:0030835	biological_process	negative regulation of actin filament depolymerization	0.021862
GO:0032412	biological_process	regulation of ion transmembrane transporter activity	0.021862
GO:0043933	biological_process	macromolecular complex subunit organization	0.021977
GO:0034622	biological_process	cellular macromolecular complex assembly	0.021977
GO:0033036	biological_process	macromolecule localization	0.022324
GO:0051649	biological_process	establishment of localization in cell	0.022772
GO:0007015	biological_process	actin filament organization	0.024076
GO:0060627	biological_process	regulation of vesicle-mediated transport	0.024674
GO:0048585	biological_process	negative regulation of response to stimulus	0.024674
GO:0031032	biological_process	actomyosin structure organization	0.024887
GO:0022898	biological_process	regulation of transmembrane transporter activity	0.024887
GO:0034765	biological_process	regulation of ion transmembrane transport	0.024887
GO:0055001	biological_process	muscle cell development	0.024887
GO:0034614	biological_process	cellular response to reactive oxygen species	0.024887
GO:0008285	biological_process	negative regulation of cell proliferation	0.025081
GO:0032965	biological_process	regulation of collagen biosynthetic process	0.026092
GO:0044253	biological_process	positive regulation of multicellular organismal metabolic process	0.026092
GO:0022409	biological_process	positive regulation of cell-cell adhesion	0.026092
GO:0007155	biological_process	cell adhesion	0.026679
GO:0022610	biological_process	biological adhesion	0.027131
GO:0051246	biological_process	regulation of protein metabolic process	0.029339
GO:0051272	biological_process	positive regulation of cellular component movement	0.02993
GO:0048583	biological_process	regulation of response to stimulus	0.030151
GO:0030334	biological_process	regulation of cell migration	0.030151
GO:0010712	biological_process	regulation of collagen metabolic process	0.030151
GO:0030834	biological_process	regulation of actin filament depolymerization	0.030151
GO:0006979	biological_process	response to oxidative stress	0.030151

GO:0043392	biological_process	negative regulation of DNA binding	0.030151
GO:0050727	biological_process	regulation of inflammatory response	0.030151
GO:0048199	biological_process	vesicle targeting, to, from or within Golgi	0.030151
GO:0051090	biological_process	regulation of transcription factor activity	0.030239
GO:0090046	biological_process	regulation of transcription regulator activity	0.030239
GO:0006119	biological_process	oxidative phosphorylation	0.030239
GO:0009206	biological_process	purine ribonucleoside triphosphate biosynthetic process	0.030239
GO:0090066	biological_process	regulation of anatomical structure size	0.031397
GO:0009145	biological_process	purine nucleoside triphosphate biosynthetic process	0.031755
GO:0009201	biological_process	ribonucleoside triphosphate biosynthetic process	0.031755
GO:0016052	biological_process	carbohydrate catabolic process	0.031755
GO:0032535	biological_process	regulation of cellular component size	0.03264
GO:0065009	biological_process	regulation of molecular function	0.033735
GO:0046164	biological_process	alcohol catabolic process	0.03425
GO:0006898	biological_process	receptor-mediated endocytosis	0.034659
GO:0009142	biological_process	nucleoside triphosphate biosynthetic process	0.035092
GO:0050865	biological_process	regulation of cell activation	0.035262
GO:0051668	biological_process	localization within membrane	0.035262
GO:0043500	biological_process	muscle adaptation	0.035262
GO:0046700	biological_process	heterocycle catabolic process	0.035954
GO:0042476	biological_process	odontogenesis	0.037093
GO:0043244	biological_process	regulation of protein complex disassembly	0.037093
GO:0031347	biological_process	regulation of defense response	0.03742
GO:0048731	biological_process	system development	0.037826
GO:0046034	biological_process	ATP metabolic process	0.038235
GO:0065007	biological_process	biological regulation	0.040556
GO:0043623	biological_process	cellular protein complex assembly	0.040581
GO:0002682	biological_process	regulation of immune system process	0.041208
GO:0019216	biological_process	regulation of lipid metabolic process	0.042217
GO:0009653	biological_process	anatomical structure morphogenesis	0.042573

GO:0010596	biological_process	negative regulation of endothelial cell migration	0.042611
GO:0051279	biological_process	regulation of release of sequestered calcium ion into cytosol	0.042611
GO:0031960	biological_process	response to corticosteroid stimulus	0.042611
GO:0043254	biological_process	regulation of protein complex assembly	0.042611
GO:0006376	biological_process	mRNA splice site selection	0.042611
GO:0030168	biological_process	platelet activation	0.042711
GO:0034762	biological_process	regulation of transmembrane transport	0.042711
GO:0006413	biological_process	translational initiation	0.042711
GO:0070482	biological_process	response to oxygen levels	0.043202
GO:0034220	biological_process	ion transmembrane transport	0.04584
GO:0030336	biological_process	negative regulation of cell migration	0.04584
GO:0006807	biological_process	nitrogen compound metabolic process	0.046188
GO:0048856	biological_process	anatomical structure development	0.046188
GO:0001937	biological_process	negative regulation of endothelial cell proliferation	0.046188
GO:0006195	biological_process	purine nucleotide catabolic process	0.046188
GO:0006006	biological_process	glucose metabolic process	0.046188
GO:0006739	biological_process	NADP metabolic process	0.046188
GO:0034621	biological_process	cellular macromolecular complex subunit organization	0.046188
GO:0060021	biological_process	palate development	0.046188
GO:0044246	biological_process	regulation of multicellular organismal metabolic process	0.046188
GO:0048145	biological_process	regulation of fibroblast proliferation	0.046188
GO:0003158	biological_process	endothelium development	0.046188
GO:0045446	biological_process	endothelial cell differentiation	0.046188
GO:0006984	biological_process	ER-nuclear signaling pathway	0.046188
GO:0000097	biological_process	sulfur amino acid biosynthetic process	0.046188
GO:0032088	biological_process	negative regulation of NF-kappaB transcription factor activity	0.046188
GO:0002698	biological_process	negative regulation of immune effector process	0.046188
GO:0051100	biological_process	negative regulation of binding	0.046252

GO:0042127	biological_process	regulation of cell proliferation	0.047353
GO:0040012	biological_process	regulation of locomotion	0.049357
GO:0007229	biological_process	integrin-mediated signaling pathway	0.049879
GO:0022626	cellular_component	cytosolic ribosome	5.92E-22
GO:0005840	cellular_component	ribosome	1.83E-18
GO:0033279	cellular_component	ribosomal subunit	3.59E-17
GO:0005829	cellular_component	cytosol	5.51E-17
GO:0044445	cellular_component	cytosolic part	3.26E-16
GO:0044444	cellular_component	cytoplasmic part	1.69E-15
GO:0005737	cellular_component	cytoplasm	1.07E-14
GO:0030529	cellular_component	ribonucleoprotein complex	2.12E-12
GO:0022627	cellular_component	cytosolic small ribosomal subunit	3.21E-11
GO:0022625	cellular_component	cytosolic large ribosomal subunit	5.10E-10
GO:0015934	cellular_component	large ribosomal subunit	1.39E-08
GO:0015935	cellular_component	small ribosomal subunit	1.39E-08
GO:0043228	cellular_component	non-membrane-bounded organelle	4.14E-07
GO:0043232	cellular_component	intracellular non-membrane-bounded organelle	4.14E-07
GO:0032991	cellular_component	macromolecular complex	6.87E-07
GO:0044424	cellular_component	intracellular part	1.32E-06
GO:0043229	cellular_component	intracellular organelle	1.29E-05
GO:0043226	cellular_component	organelle	1.45E-05
GO:0044446	cellular_component	intracellular organelle part	2.99E-05
GO:0044422	cellular_component	organelle part	3.95E-05
GO:0005622	cellular_component	intracellular	4.82E-05
GO:0000275	cellular_component	mitochondrial proton-transporting ATP synthase complex, catalytic core F(1)	0.000116
GO:0044421	cellular_component	extracellular region part	0.000163
GO:0045261	cellular_component	proton-transporting ATP synthase complex, catalytic core F(1)	0.000256
GO:0005578	cellular_component	proteinaceous extracellular matrix	0.000484
GO:0031012	cellular_component	extracellular matrix	0.001193
GO:0005753	cellular_component	mitochondrial proton-transporting ATP synthase complex	0.001228

GO:0005581	cellular_component	collagen	0.001475
GO:0045259	cellular_component	proton-transporting ATP synthase complex	0.001803
GO:0009986	cellular_component	cell surface	0.005759
GO:0031091	cellular_component	platelet alpha granule	0.014215
GO:0033178	cellular_component	proton-transporting two-sector ATPase complex, catalytic domain	0.014411
GO:0005615	cellular_component	extracellular space	0.014499
GO:0044420	cellular_component	extracellular matrix part	0.014866
GO:0016281	cellular_component	eukaryotic translation initiation factor 4F complex	0.016143
GO:0005740	cellular_component	mitochondrial envelope	0.017234
GO:0001725	cellular_component	stress fiber	0.019553
GO:0016023	cellular_component	cytoplasmic membrane-bounded vesicle	0.019975
GO:0031410	cellular_component	cytoplasmic vesicle	0.020257
GO:0031093	cellular_component	platelet alpha granule lumen	0.020799
GO:0032432	cellular_component	actin filament bundle	0.024887
GO:0031966	cellular_component	mitochondrial membrane	0.025749
GO:0044455	cellular_component	mitochondrial membrane part	0.026092
GO:0060205	cellular_component	cytoplasmic membrane-bounded vesicle lumen	0.026092
GO:0005583	cellular_component	fibrillar collagen	0.026092
GO:0030126	cellular_component	COPI vesicle coat	0.026092
GO:0031988	cellular_component	membrane-bounded vesicle	0.026522
GO:0042641	cellular_component	actomyosin	0.028018
GO:0016469	cellular_component	proton-transporting two-sector ATPase complex	0.028584
GO:0031982	cellular_component	vesicle	0.030151
GO:0031983	cellular_component	vesicle lumen	0.030151
GO:0032587	cellular_component	ruffle membrane	0.030151
GO:0030663	cellular_component	COPI coated vesicle membrane	0.030151
GO:0030131	cellular_component	clathrin adaptor complex	0.031321
GO:0031090	cellular_component	organelle membrane	0.034106
GO:0030119	cellular_component	AP-type membrane coat adaptor complex	0.035092

GO:0016581	cellular_component	NuRD complex	0.035262
GO:0034362	cellular_component	low-density lipoprotein particle	0.035262
GO:0000323	cellular_component	lytic vacuole	0.037826
GO:0005764	cellular_component	lysosome	0.037826
GO:0005576	cellular_component	extracellular region	0.04241
GO:0005773	cellular_component	vacuole	0.042611
GO:0030141	cellular_component	secretory granule	0.04584
GO:0005743	cellular_component	mitochondrial inner membrane	0.046188
GO:0030137	cellular_component	COPI-coated vesicle	0.046188
GO:0003735	molecular_function	structural constituent of ribosome	4.58E-20
GO:0005515	molecular_function	protein binding	2.08E-16
GO:0005198	molecular_function	structural molecule activity	3.75E-14
GO:0005488	molecular_function	binding	4.87E-08
GO:0048407	molecular_function	platelet-derived growth factor binding	8.56E-07
GO:0019838	molecular_function	growth factor binding	1.27E-06
GO:0005178	molecular_function	integrin binding	2.77E-05
GO:0032403	molecular_function	protein complex binding	3.48E-05
GO:0003723	molecular_function	RNA binding	6.89E-05
GO:0005102	molecular_function	receptor binding	7.38E-05
GO:0019966	molecular_function	interleukin-1 binding	0.001302
GO:0019899	molecular_function	enzyme binding	0.002052
GO:0005520	molecular_function	insulin-like growth factor binding	0.003332
GO:0046933	molecular_function	hydrogen ion transporting ATP synthase activity, rotational mechanism	0.004771
GO:0019843	molecular_function	rRNA binding	0.004771
GO:0047485	molecular_function	protein N-terminus binding	0.009769
GO:0019865	molecular_function	immunoglobulin binding	0.009941
GO:0043499	molecular_function	eukaryotic cell surface binding	0.009941
GO:0008092	molecular_function	cytoskeletal protein binding	0.013889
GO:0043531	molecular_function	ADP binding	0.014411
GO:0046961	molecular_function	proton-transporting ATPase activity, rotational mechanism	0.014411
GO:0019829	molecular_function	cation-transporting ATPase activity	0.017281

GO:0042288	molecular_function	MHC class I protein binding	0.026092
GO:0005161	molecular_function	platelet-derived growth factor receptor binding	0.026092
GO:0043498	molecular_function	cell surface binding	0.035092
GO:0051117	molecular_function	ATPase binding	0.035262
GO:0008430	molecular_function	selenium binding	0.038854
GO:0005509	molecular_function	calcium ion binding	0.041005
GO:0003743	molecular_function	translation initiation factor activity	0.042611
GO:0042803	molecular_function	protein homodimerization activity	0.046188